

Alok Hotta

☎ +91-9692604521 ✉ alohotta10@gmail.com 🔗 linkedin.com/in/alok-hotta 🐙 github.com/iamalok123 🌐 [portfolio](#)

SUMMARY

Full-Stack Software Engineer (B.Tech CSE '27) with production experience building AI-powered web applications using **React, Next.js, Node.js, TypeScript, and PostgreSQL**; shipped multiple full-stack projects integrating **LLM APIs, RAG pipelines, and background job automation**. **LeetCode Knight** — Top 5% globally, 700+ problems solved.

TECHNICAL SKILLS

Languages: C, C++, Python, JavaScript, TypeScript, SQL
Frontend: React.js, Next.js, Tailwind CSS, Redux Toolkit, Zustand, SSR, CSR
Backend: Node.js, Express.js, REST APIs, WebSocket, Socket.IO, JWT, OAuth, Zod
Databases & ORM: MongoDB, MySQL, PostgreSQL, Prisma, Drizzle, Cloudinary
AI / GenAI: RAG Pipeline, LLM APIs (Gemini, OpenAI, Claude, Groq), OpenRouter, Prompt Engineering
Tools & DevOps: Git, GitHub, Docker, Vercel, Postman, npm
Core Subjects: Object-Oriented Programming (OOPs), Database Management Systems (DBMS), Operating Systems (OS), Computer Networks (CN)

PROJECTS

Zephyr | *Next.js, TypeScript, Prisma ORM, PostgreSQL, Clerk, Arcjet, Gemini API, Sandpack* [GitHub](#) | [Live](#)

- Built an **AI website builder** using **Next.js 16** and **Gemini API**, streaming real-time AI thought process via **SSE** with instant in-browser preview and automated **npm package verification** to resolve hallucinated imports against the live registry.
- Integrated a **Sandpack-based sandbox** for live client-side code editing and hot-reload preview, removing server-side execution overhead.
- Secured API routes with **Clerk** authentication and **Arcjet** prompt-injection detection and rate limiting, blocking malicious payloads.
- Used atomic **Prisma transactions** to deduct credits and persist state, preventing charges on failed generations.

StudyFlow | *React.js, TypeScript, Express.js, MongoDB, Gemini API, Cloudinary, OAuth* [GitHub](#) | [Live](#)

- Architected an end-to-end **RAG pipeline** using Express.js and Gemini API with **semantic chunking** and keyword-based retrieval to power context-aware summaries, quizzes, and flashcards.
- Built multi-document **Workspaces** with custom theming to unify **AI Chat, Mind Maps**, summaries, flashcards, and quizzes for grouped document sets.
- Added real-time **Gemini response streaming** and interactive **Mind Maps** (React Flow) for documents and workspaces, plus **Markdown rendering** (code blocks, tables, lists) for AI answers.
- Implemented route-level **lazy loading** (**40%** faster initial load) and secured auth via Google **OAuth/JWT**.

Social AI | *TypeScript, Node.js, Express.js, MongoDB, Gemini API, Cloudinary, Cloudflare* [GitHub](#) | [Live](#)

- Built a **full-stack social scheduler** with a dual **AI engine** using **Gemini API** for captions and content ideas.
- Engineered a **2-tier image-generation fallback** (Cloudflare Workers AI, Pollinations.ai) uploaded to **Cloudinary**.
- Automated **post publishing** (Twitter/X, LinkedIn, Facebook, Instagram) via **node-cron** with real-time activity logging.
- Secured the **REST API** with **JWT**, Google **OAuth 2.0**, and **bcrypt**; built an interactive **Kanban board** for content ideation.

ACHIEVEMENTS

- **LeetCode - Knight:** Top 5% globally, 600+ problems solved in Data Structures and Algorithms (DSA).
- **GeeksforGeeks:** Institute Rank #1, 500+ problems solved in DSA.
- **Smart India Hackathon 2025:** Led a 6-member team to design and present a **full-stack AI automation MVP**, securing **1st place** among 50+ teams at the internal SIH round (Ministry of Power, Govt. of India).

EDUCATION

Gandhi Engineering College

Bachelor of Technology in Computer Science Engineering; CGPA: 8.9/10.0

Bhubaneswar, Odisha

2023 – 2027